An Oriented Ship Detection Method of Remote Sensing Image With Contextual Global Attention Mechanism and Lightweight Task-Specific Context Decoupling

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***Abstract*— Ship detection in remote sensing images has been attracting a lot of attention due to its great application value in both military and civilian fields. However, ships in high-resolution remote sensing images are characterized by the remarkable features of multiscale, arbitrary orientation, and dense arrangement, which is a great challenge for fast and accurate target detection. In order to solve problems, we propose a YOLOV5-based oriented ship detection method of remote sensing images with contextual global attention mechanism and lightweight task-specific context decoupling (CGTC-RYOLO) in this article. First, a cross-stage partial context transformer (CSP-COT) module is introduced to capture global contextual spatial relations using multihead self-attention (MHSA) to verify their implications in implicit dependencies. Second, we propose an angle classification prediction branch in the YOLOV5 head network for detecting targets in any direction and design probability and distribution loss function (PrfoIoU) to optimize the regression effect. Third, the lightweight task-specific context decoupling (LTSCODE) for target detection is employed to replace the original head in the YOLOV5 model, which is used to solve the accuracy problem caused by YOLOV5's hybridization of classification and localization. Ablation experiments demonstrate the importance and effectiveness of each module. Compared with the benchmark model, the CGTC-RYOLO has the 5.9%, 3.7%, and 4.3% mAP improvements on the DOTA-ship dataset, the HRSC2016 dataset, and the UCAS-AOD dataset, respectively. Moreover, the model's generalization is also validated. Compared with state-of-the-art methods, the CGTC-RYOLO can achieve better accuracy and fewer parameters.**

***Index Terms*— Decoupling, probability and distribution loss function (PrfoIoU), rotation detection, target detection, transformer, you only look once (YOLO).**

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I. Introduction

HORIZONTAL bounding box (HBB) detection has been widely used in conventional natural scene target detection [1], [2], [3], [4], [5]. One- and two-stage detectors are two different categories into which previous work on HBB models can be classified. One-stage detectors are for simultaneous target location localization and target category identification, and only one feature extraction is needed to detect the target. The two-stage detector is a method that sequentially performs both localization and recognition detection. It is noteworthy that one-stage methods are widely used in visual image detection. This is because they are characterized by a small number of parameters and an acceptable detection accuracy.

The popular method of a one-stage detector is you only look once (YOLO) series. Among them, YOLOV1 [6] serves as the initial detector of the series. YOLOV1's network architecture, which has two fully connected layers and 24 convolutional layers, is based on GoogLeNet [7]. Compared to YOLOV1, YOLOV2 stabilizes the training by predicting the presence of the same number of categories and objects as the number of anchor boxes. YOLOV3 [8] predicts three different proportions of boxes and extracts the features in a similar way to a feature pyramid network (FPN). The main improvement of YOLOV4 [9] suggests optimizing the original backbone network with an improved architecture that integrates bag-of-specials [10]. YOLOV5 proposes different cross-stage partial (CSP) backbone structures for the backbone and neck, which strengthens the capability of network feature fusion and solves the problem of repeated gradients in other large convolutional network structures. Later, YOLOV6 [11] and YOLOVX [12] were proposed. These two models are improved by using an anchor-free frame allocation strategy and the new decoupling head method. However, YOLOV5 is still the most widely

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used first-stage algorithm for object detection due to its lightweight model size, fast training speed, mature framework system, and higher object detection speed. There are four versions of YOLOV5: YOLOV5s, YOLOV5m, YOLOV5l, and YOLOV5x. Among the YOLOV5 series, YOLOV5s are the network with the smallest feature map width and depth. To reduce the number of parameters, we choose YOLOV5s as our baseline model. All the following discussion about YOLOV5 refers to YOLOV5s.

However, ships in high-resolution remote sensing images are characterized by the remarkable features of multiscale, arbitrary orientation, and dense arrangement, which is a great challenge for fast and accurate target detection [13], [14]. Existing detectors often suffer from imbalances between classification and regression tasks, incomplete rotation frameworks, and ineffective use of contextual global relationship modeling. Therefore, this article proposed a YOLOV5-based oriented ship detection method of remote sensing images with contextual global attention mechanism and lightweight task-specific context decoupling (CGTC-RYOLO). Overall, the contributions of this article are given as follows.

1) An oriented target detection frame of remote sensing image is proposed. By designing a probability and distribution loss function (PrfoIoU) to categorize angles, the rotation detection of targets can be fast and accurately realized.
2) A new lightweight task-specific context decoupling (LTSCODE) head is proposed. The LTSCODE pays more attention to the fusion of contextual features and solves the problem of accuracy degradation caused by the imbalance of tasks in the head part.
3) A contextual CSP and contextual transformer (CSP-COT) module is proposed in the detection framework. The model can effectively combine the spatial context and channel information fusion mechanism based on the COT structure with the channel-based information fusion mechanism based on the cross-stage partial network (CSPNet). This can not only alleviate the problem of the underlying semantic graph becoming less associated with the initial feature graph in space and channel due to the depth of the CSPNet but also strengthen the information fusion mechanism of the CSP structure to realize efficient fusion of spatial and channel information.
4) Numerous results on the DOTA dataset, the HRSC2016 dataset, and the UCAS-AOD dataset demonstrate that the proposed method has accurate detection capability and excellent generalization ability [66], which outperforms other advanced methods.

The rest of the article is organized as follows. Section II introduces the relevant works of this article. Section III describes the proposed target detection model in detail. Section IV describes and analyzes the experimental results. Section V concludes this study.

II. Related Works

In view of the advantages of YOLOV5, many works have been made in target detection of remote sensing images. Several representative examples are given as follows. Zhang et al. [15] propose a new lightweight object detection algorithm based on YOLOV5s to improve detection performance while ensuring fast processing and wide applicability. In order to recognize targets with varying scales, Dai et al. [16] devised a technique that adjusts the feature maps' output layer, crops the image, and uses a convolutional neural network. This method is appropriate for high-resolution remote sensing images for object detection. Cao et al. [17] proposed a lightweight YOLOV5 network, which introduced a lightweight module and a new small-target detection head to improve detection accuracy and reduce parameters. In summary, since most of the objects in remote sensing images have the characteristics of multiscale, arbitrary angle, and dense arrangement, three major challenges have been brought to remote sensing image detection based on YOLOV5.

1) *The First Is the Imbalance Between the Classification and Regression Tasks of the Predictor Header:* The current problem is that the mainstream methods use the same features for classifying and locating branches in the head section. Representative methods, such as the YOLO family before V5, use a 1×1 convolution kernel as the decoupling head, leading to not only not distinguishing between localization and classification branches but also not providing different features for classification and localization branches to use. To solve this problem, several attempts have already been made. Ge et al. [12] argued that classification and localization, two different tasks, need to be performed separately to obtain better performance, so they proposed the decoupled head structure. Then, Li et al. [11] proposed a lightweight decoupling head improvement method, which can obtain accurate accuracy while reducing the number of parameters. Dai et al. [18] proposed a new dynamic head framework that uses an attention mechanism to unify different target detection heads. However, these methods still cannot solve the problem that the head part of the model still uses the same features. The problem is that by combining the classification and localization tasks together or by decomposing the two tasks only at the parameter level, the head network still obtains feature information from shared features. This can lead to an imbalance between the two tasks of classification and localization, i.e., a bias toward one task, thereby inhibiting the other [19], [20]. Although the above method solves part of the problems of classification and localization to some extent, such decoupling heads still have the intractable deep contradiction of sharing the same features for classification and localization.
2) *The Second Is the Problem of Rotating Frame:* While HBB detectors have shown remarkable success in natural situations, performance degrades significantly due to the objects rotating and densely packed when these detectors are used directly in remote sensing images. This is because the efficiency of the HBB is low when the direction of the object is not aligned with the coordinate axis. Meanwhile, there will be a large area of white

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space and false crosses, resulting in the failure to provide accurate rotation and scale information. In this case, the angle prediction method can be introduced into the oriented border box (OBB) detection to obtain the predicted value of the output angle, so as to obtain more accurate regions and reduce the background noise. Ma et al. [21] proposed an additional branch of angle prediction and added it to the Faster RCNN to detect the oriented objects. High-quality anchors were selected by Ming et al. [22] in order to reduce the imbalance between the tasks of regression and classification. Yang and Yan [23] designed a circular smooth labeling (CSL) technique to analyze rotated angles. Ding et al. [24] proposed a region of interest (ROI) technique that converts a horizontal ROI into a rotated ROI. In general, the current remote sensing image detection methods at any azimuth still lack accurate and lightweight structure and real-time modeling.

3) *The Third Is the Problem of Using Contextual Global Relationship Modeling Effectively:* Attention mechanism is very important for global relationship modeling. Multihead self-attention (MHSA) mechanism [25], [26] can not only effectively learn the relationship between distant objects [27], [28] but also effectively filter out noise and some occlusive interference [29]. As a result, a lot of studies have focused on the mechanism of attention. Zhu et al. [30] introduced MHSA into the backbone network and detection head of YOLO. Li et al. [31] designed an MHSA that makes full use of contextual information. Srinivas et al. [32] proposed a new bottleneck module by replacing 3×3 convolution with the MHSA module. Elfwing et al. [33] proposed a YOLO model that combines a channel attention mechanism to detect small objects in aerial images. Wang et al. [34] proposed a YOLO model, adding a small target detection head and a new transformer module to realize the detection of small targets in aerial images. Ding et al. [24] proposed a YOLO model that introduces a new transformer module and a feature fusion module for the accurate detection of remote sensing images. Vaswani et al. [25] proposed a YOLO framework that fuses a new bidirectional feature fusion module (bi-DFM) for detecting ships in high-resolution synthetic aperture radar imagery. Ramachandran et al. [27] introduced an attention module fused with a transformer module to detect rotating objects. All these works pay much attention to the MHSA but ignore the two parts of information hidden in the input characteristics: static context information and dynamic context information.

Especially in the object detection of remote sensing images, the object to be detected often has a strong correlation in spatial, and the feature extraction of static and dynamic context information can reflect the potential spatial distribution relationship in the image [35]. Therefore, the context information is very important for target detection.

In summary, the above attempts can inspire us in practical applications as follows.

1) *In Terms of the Imbalance:* To make the classification and regression tasks balanced, an effective strategy is to give specific feature maps for specific tasks. Localization in the model requires more boundary-aware features to accurately regress the bounding box, while object classification requires more semantic contextual information to accurately classify them. Thus, this inspires us that feature maps that incorporate contextual information for different tasks should be provided.
2) *In Terms of the Rotating Frame:* Designing a PrfoIoU to solve the problem of rotating frames, which designs the detection algorithm in a more semantic segmentation-oriented computational form to enhance the accuracy of localization. This method can make angle regression more efficient and accurate.
3) *In Terms of the Context Global Relationship Modeling:* The spatial transformer module can be used to enhance the efficiency of extracting context global information. The self-attention mechanism in the transformer can extract rich contextual global information and improve contextual global relationship modeling capabilities [36].

Considering the low parameters, high-performance characteristics of YOLOV5, and the above inspirations, this article proposes a CGTC-RYOLO model to deal with the above challenges.

III. Model Expressions

First, we present the general structure for target detection in remote sensing images and then present some specific improvements, including CSP-COT, LTSCODE, PrfoIoU, and an improved multitask loss function.

A. Overview of the Proposed Model

1) Model Structure: The general structure of the network is shown in Fig. 1. The modified model is divided into three parts: backbone, neck, and prediction head.

a) Backbone: The backbone includes four C3 modules and one spatial pyramid pooling–fast (SPPF) module. Among them, the C3 module is the abbreviation of three CSP bottlenecks, and the structure of the CSP bottleneck is shown in Fig. 2(b). The main function of SPPF is to improve the fusion of multiple receptive fields and the detection performance of small targets. In order to strengthen the backbone's ability to extract context global relationships and reduce computation load, we replace the CSP bottleneck in the fourth C3 module with CSP-COT.

b) Neck: The path aggregation network (PAN) and FPN constitute the neck. Just as the reason for adding CSP-COT in the backbone, we replaced the CSP bottleneck with CSP-COT in the first C3 module of the neck. Since each LTSCODE requires three layers of feature maps as input, we add an extra layer of C3 module at the end of the neck part to obtain a feature map P_6 for LTSCODE.

c) Head: We replaced the original head with the new LTSCODE because the original head will cause an imbalance between the two tasks of classification and positioning.

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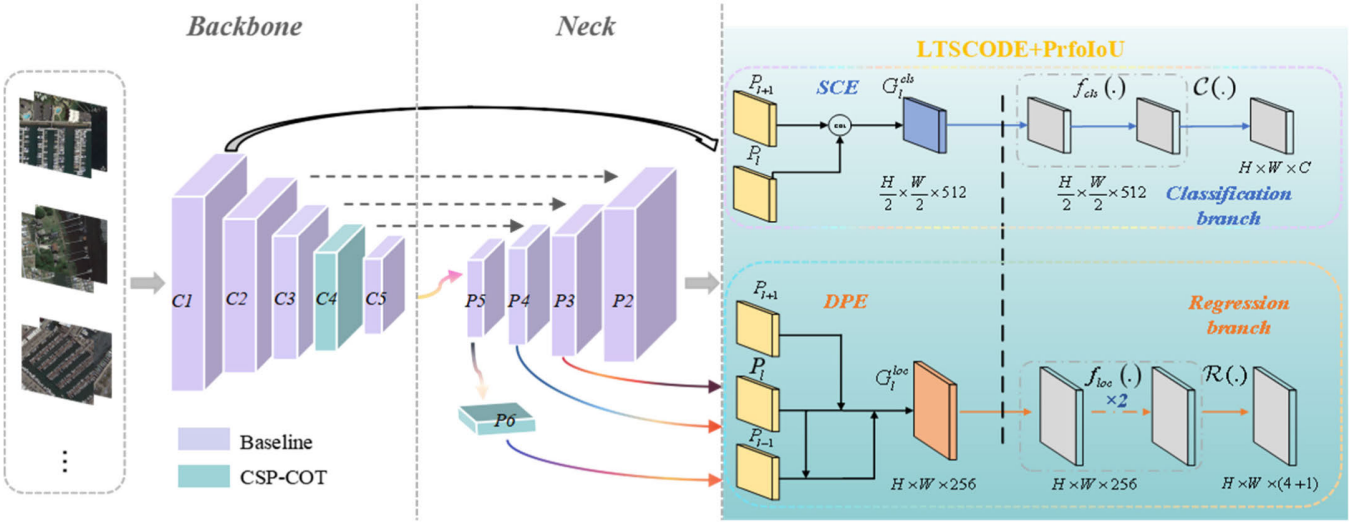


Fig. 1. General structure of our proposed network. A backbone of YOLOV5 is used, a CSP-COT is used at the end of the backbone network, the neck part is a combined structure of FPN and PANet, a CSP-COT is also used at the beginning of the neck, and finally, an LTSCODE decoupled head structure and an angle categorization prediction branch are added to the prediction head part.

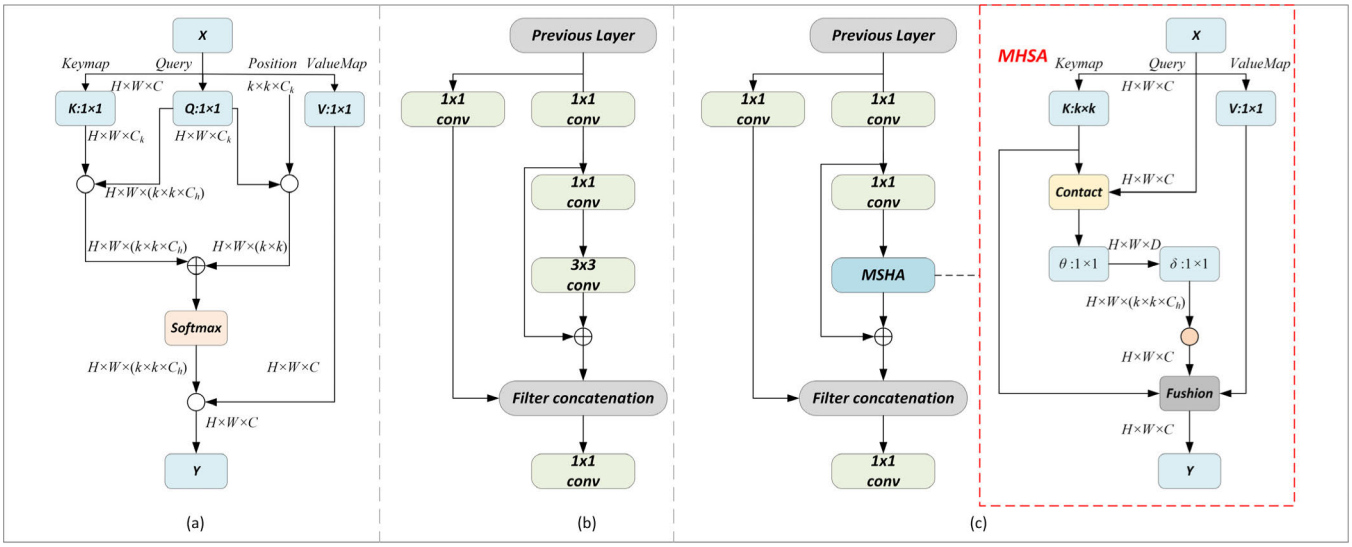


Fig. 2. CSP bottlenecks and MHSA. (a) Conventional self-attention. (b) CSP bottleneck. (c) CSP context Transformer. ⊕ and ⊗ denote the elementwise sum and local matrix multiplication.

LTSCODE has a classification branch, a regression branch, and an angle branch. The classification branch uses semantic context encoding (SCE), and the regression branch uses detail-preserving encoding (DPE).

2) Algorithmic Process: We first feed the image into the backbone network and then extract the main features through the backbone network. The neck structure then uses the PAN to deliver the high-level semantic information to the low-level feature map and the FPN module to transfer the low-level feature information to the high-level feature map. In this process, a richer context global feature map can be obtained through the CSP-COT module. Finally, the fusion of contextual information and features is enhanced by the TSCODE module, which provides the required information for both classification and regression tasks at different stages to perform the corresponding operations. Multitask losses are computed in the training stage, including bounding box and angular regression loss and classification loss. During testing, nonmaximization suppression (NMS) is used to build the detection predictions.

In summary, Algorithm 1 illustrates the training process of the CGTC-RYOLO.

B. CSP-COT

The CSPNet structure can not only optimize the gradient but also reduce the computational cost [37]. The CSP bottleneck designed by the CSPNet structure has become a basic component of YOLOV5, as shown in Fig. 2(b). We integrate MHSA into the CSP structure, use it to replace the 3 × 30 convolution, and propose a CSP-COT, as shown in Fig. 2(c).

The CSP-COT module is a hybrid of convolution and MHSA, where the self-attention mechanism is the most important component. The most common traditional self-attention mechanism is shown in Fig. 2(a). The size of the input feature map X is $H \times W \times C$ (H: height; W: width;

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Algorithm 1 CGTC-RYOLO Training

Input: I: input images;
ξ: learning rate;
α, β: trade-off hyperparameter;
$\mathcal{t}$: sample batch;
∇_η: gradient
Output: model parameter η;
randomly initialize η;
repeat
  for each task $\mathcal{t}_i \in \mathcal{t}$ **do**
    sample $\{I\} \sim \mathcal{t}_i$;
    Backbone: calculate input image $I \rightarrow$ feature maps: C_1, C_2, C_3, C_4, C_5;
    Neck: using CSP-COT and feature map $C_3, C_4, C_5 \rightarrow$ representative feature maps: P_3, P_4, P_5, P_6;
    for P_i in $[P_3, P_4, P_5]$ **do**
      SCE: $Concat(DConc(P_l), P_{l+1}) \rightarrow$ obtain semantic feature maps G_l^{cls}, $\mathcal{F}_c(G_l^{cls}) \rightarrow$ classification predictions $P_{cls} \rightarrow L_{cls}$;
      DPE: use $P_{l-1}, P_l, P_{l+1} \rightarrow$ obtain spatial feature maps G_l^{loc}, $\mathcal{F}_c(G_l^{loc}) \rightarrow$ location predictions $P_{loc+angle} \rightarrow L_{loc+angle}$
Loss function: $L = \alpha L_{loc+angle} + \beta L_{cls}$;
    Update model: $\eta \leftarrow \eta - \xi * \nabla_\eta L$;
  end for;
end for;
until convergence;

and C: channel number). X is then transformed by embedding the matrix (W_q, W_k, W_v) into the queries $Q = XW_q$, keys $K = XW_k$, and values $V = XW_v$, where C_h represents the number of heads, $\otimes$ stands for matrix multiplication, k is constant, and Y is the concatenation of aggregated feature maps for all heads.

Our MHSA modules are different from traditional modules. Its main purpose is to enhance self-attention to static context information. The MHSA utilizes the context information between neighboring keywords to promote self-attentional learning in an efficient way and enhances the representation power of the output aggregate feature map. Assuming that the input is a 2-D feature map X of size $H \times W \times C$, keys, queries, and values are defined respectively as $K = X$, $Q = X$, and $V = XW_v$. In order to contextualize each key representation, the COT block first uses $k \times k$ group convolution over all of the neighbor keys within the $k \times k$ grid spatially. The contextual information among local neighbor keys is naturally reflected in the learned contextualized keys $K^1 \in \mathbb{R}^{H \times W \times C}$. Then, we use K^1 as the static context representation of input X. Next, two sequential 1×1 convolutions (W_θ with ReLU activation function and W_δ without activation function) are used to achieve the attention matrix, which is conditioned on the concatenation of contextualized keys K^1 and queries Q

$$A_2 = \left[K^1, Q\right] W_\theta W_\delta. \tag{1}$$

For each head, the local attention matrix at each spatial location of A_2 is learned based on the query feature and the contextualized key feature. By using the mined static context K^1 as extra direction, this method improves self-attention learning. Next, we aggregate all values V to calculate the attended feature map K^2 based on the contextualized attention matrix A_2

$$K^2 = V \otimes A_2. \tag{2}$$

The resulting output Y is a blend of both the static context K^1 and the dynamic context K^2.

We add CSP-COT at the end of the backbone and at the beginning of the PANet. There are two main reasons for this. First, these two stages contain more semantic information and lower feature map resolution, and the computation and storage overhead can be reduced. Using the CSP-COT module at this stage can effectively capture global information and a lot of context information. Second, as the network layers deepen, the channel and spatial information of the feature map becomes scarce and less relevant to the initial feature map. By exploiting the ability of CSP-COT to extract spatial information, we can effectively mitigate the attenuation of spatial and channel information in the network to produce a more comprehensive feature representation. This integration effectively captures the dependencies of spatial information and improves the overall discriminative ability of the model.

Compared to the original CSP module, CSP-COT reduces the number of parameters and obtains more information. We will demonstrate in the experimental section in the following that integrating the CSP-COT module in the YOLOV5 framework is superior to the other module.

C. LTSCODE

In object detection, classification and localization are two closely related but incompatible tasks. For each target, classification requires richer contextual semantic information, while localization requires more detailed high-resolution boundary information. In YOLOV5, both the classification task and the localization task use 1×1 convolution for learning. Some investigations [11], [12] have introduced different decoupled head structures to alleviate the highly correlated but contradictory problem of classification and positioning tasks. However, these structures cannot completely solve the problem because the semantic and contextual information of the inputs seriously rely on the shared input features P_l. The common input features greatly limit the training of the two branches of classification and regression.

In order to solve this problem in YOLOV5, we introduced TSCODE into the YOLOV5 framework and made corresponding lightweight improvements to make it more adaptable to the YOLOV5 framework. Specifically, the number of convolutional layers is set to 1 in $f_{\text{cls}}(\cdot)$ and 2 in $f_{\text{loc}}(\cdot)$ without drastically affecting accuracy.

TSCODE separates the feature encoding of the two tasks on the source code and uses feature maps with different semantic contexts in the two branches. Instead of using P_l as the common input, we provide task-specific input features for both

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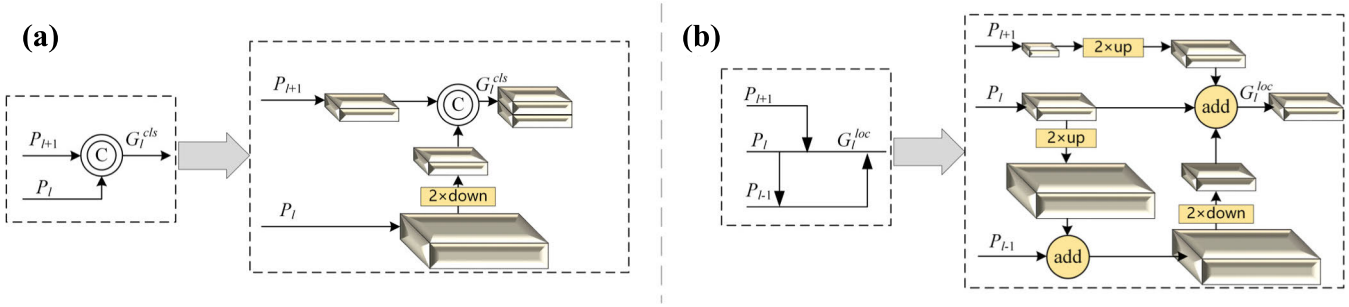


Fig. 3. Two basic structures in LTSCODE. (a) SCE module. (b) DPE module.

branches, namely, G_l^{cls} and G_l^{loc}. As a result

$$\mathcal{L} = \gamma \mathcal{L}_{\text{cls}}\left(\mathcal{F}_c\left(G_l^{\text{cls}}\right), C\right) + \beta \mathcal{L}_{\text{loc}}\left(\mathcal{F}_r\left(G_l^{\text{loc}}\right), \mathcal{B}\right) \tag{3}$$

where β and γ are tradeoff hyperparameters. For the classification branch, we generate spatially coarser but semantically richer feature maps. For the localization branch, we provide feature maps that contain more detailed texture and boundary information.

1) Semantic Context Encoding (SCE) Module: For the classification branch, the SCE is proposed in Fig. 3. In classification tasks, the deeper semantic information is more important than the slightly shallower image information. Based on this consideration, we provide SCE with two levels of feature maps (P_l, P_{l+1}) in each layer, which is used to generate high-level semantically informative feature maps. As shown in Fig. 3, we first downsample P_l by a factor of 2 and subsequently concatenate it with P_{l+1} to generate G_l^{cls}

$$G_l^{\text{cls}} = \text{Concat}(\text{DConv}(P_l), P_{l+1}) \tag{4}$$

where Concat$(\cdot)$ and DConv$(\cdot)$ denote the cascaded and shared downsampled convolutional layers in the SCE module, G_l^{cls}, at half the resolution of P_l and subsequently converted to $\mathcal{F}c(\cdot) = \{f_{\text{cls}}(\cdot), \mathcal{C}(\cdot)\}$ to predict classification scores. In this way, we can not only utilize the sparsity of salient features (from P_l) but also benefit from the rich semantic context at higher pyramid levels (from P_{l+1}). This helps to infer object classes more efficiently, especially for objects that are not textured or have heavy occlusions.

2) Detail-Preserving Encoding (DPE) Module: Unlike classification, localization is a more fine-grained task that relies on more texture details and boundary information to predict the corner points of an object. However, existing methods typically regress the corner points of an object from a single-scale feature map. Lower pyramid-level feature maps are more responsive to the contours, edges, and detailed textures of the object. This can further help the localization task but usually comes with significant additional computational cost [38], [39]. In addition, advanced feature maps are also important for positioning because they help to see the whole object as completely as possible, which provides more information for inferring the overall shape of the object. Combining these aspects, the DPE is proposed in Fig. 3, where we provide three levels of feature maps (P_{l-1}, P_l, and P_{l+1}) to the DPE in each layer. P_{l-1} provides more details and edge features, and P_{l+1} extends the sensory field to obtain more comprehensive macroscopic information. P_l is upsampled twice, followed by an add operation with P_{l-1}, and then, its resolution is downsampled to P_l by a 3×3 convolution with a step size of 2. Finally, P_{l+1} is upsampled and added to obtain the final G_l^{loc}. The formula is given as follows:

$$G_l^{\text{loc}} = P_l + \mu(P_{l+1}) + \text{DConv}(\mu(P_l) + P_{l-1}) \tag{5}$$

where $\mu(\cdot)$ stands for upsampling and DConv$(\cdot)$ represents a shared downsampling convolution layer. We use the second feature layer extracted by the network C_2 instead of the feature map P_2 to compute G_3^{loc} together with P_3 and P_4. Since P_2 computed via FPN incurs significant computational cost, G_l^{loc} is subsequently converted to $\mathcal{F}r(\cdot) = \{f_{\text{loc}}(\cdot), \mathcal{R}(\cdot)\}$ to predict the bounding box and confidence.

D. Target Boundary Processing at Any Angle

Most of the targets in remote sensing images are dense, complex, and nondirectional, the method of using horizontal bounding box regression is no longer applicable, and we need to correct these rotating targets in order to more accurately detect objects at any angle.

In this article, we have designed a new angle loss function PrfoIoU, which is based on the design concept that semantic segmentation tasks can accurately locate the shape of the target. If the detection task is transformed into a semantic segmentation task, the angle problem will be solved.

The traditional form of object detection box is mostly (x, y, h, w, θ), and the detection box form of PrfoIoU is designed as an elliptical Gaussian distribution detection box, which is closer to the semantic segmentation shape of the target and better reflects the positioning of the target. After that, PrfoIoU uses the Bhattacharyya coefficient (BC) to calculate the degree of overlap between the target box and the real box. The BC between the two probability density functions $p(x)$ and $q(x)$ is defined as follows:

$$\text{BC}(p, q) = \int R^2 \sqrt{p(x)q(x)dx} \tag{6}$$

where $\text{BC}(p, q) \in [0, 1]$, and when the two distributions are the same, $\text{BC}(p, q) = 1$.

Based on the above formula, the Babbitt distance (BD) between different distributions can be obtained

$$\text{BD}(p, q) = -\ln \text{BC}(p, q). \tag{7}$$

From the above two equations, it can be seen that BD is negatively correlated with BC, where BD represents an

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estimate of the similarity between p and q. Assuming that the Gaussian distribution of two rotating boxes is $p \sim N(\mu 1, \Sigma 1)$, $q \sim N(\mu 2, \Sigma 2)$

$$\mu 1 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \Sigma 1 = \begin{bmatrix} a_1 & c_1 \\ c_1 & b_1 \end{bmatrix}, \quad \mu 2 = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$$
$$\Sigma 2 = \begin{bmatrix} a_2 & c_2 \\ c_2 & b_2 \end{bmatrix}. \tag{8}$$

By combining the above equation, the closed-form expression of BD can be obtained

$$\mathrm{BD} = \frac{1}{8}(\mu 1 - \mu 2)^{\mathrm{T}} \Sigma^{-1} (\mu 1 - \mu 2) + \frac{1}{2} \ln \left(\frac{\det \Sigma}{\sqrt{\det \Sigma 1 \det \Sigma 2}} \right) \tag{9}$$

$$\Sigma = \frac{1}{2}(\Sigma 1 + \Sigma 2). \tag{10}$$

Introduce $\mu 1$ and $\mu 2$ into BD, and divide BD into B1 and B2 based on $\mathrm{BD} = B1 + B2$ and $\mathrm{BC} = e^{-\mathrm{BD}}$

$$B1 = \frac{1}{4} \frac{(a_1 + a_2)(y_1 - y_2)^2 + (b_1 + b_2)(x_1 - x_2)^2}{(a_1 + a_2)(b_1 + b_2) - (c_1 + c_2)^2} + \frac{1}{2} \frac{(c_1 + c_2)(x_2 - x_1)(y_1 - y_2)}{(a_1 + a_2)(b_1 + b_2) - (c_1 + c_2)^2} \tag{11}$$

$$B2 = \frac{1}{2} \ln \left(\frac{(a_1 + a_2)(b_1 + b_2) - (c_1 + c_2)^2}{4\sqrt{(a_1 b_1 - c_1^2)}\sqrt{(a_2 b_2 - c_2^2)}} \right). \tag{12}$$

Due to the fact that BD does not satisfy the triangle inequality, it is not a true distance. Therefore, to represent the true distance, the Hellinger distance (HD) is used, and its formula is given as follows:

$$\mathrm{HD}(p, q) = \sqrt{1 - \mathrm{BC}(p, q)} \tag{13}$$

where $\mathrm{HD}(p, q) \in [0, 1]$; assuming that the predicted Gaussian bounding box (GBB) is $p = (x_1, y_1, a_1, b_1, c_1)$ and the true GBB is $q = (x_2, y_2, a_2, b_2, c_2)$, the loss function is given as follows:

$$L_1(p, q) = \mathrm{HD}(p, q) \in [0, 1] \tag{14}$$

$$L_2(p, q) = \mathrm{BD}(p, q) = -\ln\left(1 - L_1^2(p, q)\right) \in [0, \infty]. \tag{15}$$

If $L_1(p, q)$ is used to calculate the loss, an excessive distance between the two Gaussian distributions will cause slow convergence during network training. In addition, $L_2(p, q)$ cannot well represent the relationship with IoU. Therefore, our training mode is that we will originally train with $L_2(p, q)$ and after that using $L_1(p, q)$ as the distance gets closer and closer.

It is worth noting that the direction of GBB inherits from the ellipse, and when the detection target is a square, the direction of the rotating bounding box cannot be determined. To address this issue, we introduced the distribution focal loss (DFL), ultimately forming the complete PrfoIoU loss function. The specific principle of DFL is to discretize the angle using a predefined uniform interval δ and integrate it to derive the predicted angle value θ

$$\theta = \sum_{i=0}^{90} p_i^{i\delta} \tag{16}$$

where p_i represents the probability that the angle decreases in each interval.

E. Multitask Loss Function

Based on the above considerations, an angle classification branch is added to the YOLOV5 prediction header, and the angle information is decoupled from the bounding box regression parameters, so the total loss of the network can be defined by

$$L = \alpha L_{\mathrm{loc+angle}} + \beta L_{\mathrm{cls}} \tag{17}$$

where $L_{\mathrm{loc+angle}}$ represents bounding and angle regression loss and L_{cls} represents classification loss. α and β are hyperparameters controlling the weights of the losses.

1) Angle and Loc Loss: Angle loss and loc loss are both calculated using the ProfIoU loss function proposed in this article, with specific formulas provided in Section III-D.

2) Classification Loss: The BCE loss is used to calculate classification loss

$$L_{\mathrm{cls}} = \sum_{i}^{N_p} \mathrm{BCEWithLogits}(P_{\mathrm{cls}}, T_{\mathrm{cls}}) \tag{18}$$

where P_{cls} denotes the predicted probability distribution for all categories and T_{cls} denotes the probability distribution corresponding to the truth value; besides, BCEWithLogits is defined as

$$\mathrm{BCEWitlLogits} = -\sum_{n=1}^{N} \left[y_i \times \log\sigma(x_i) + (1 - y_i) \times \log(1 - \sigma(x_i)) \right] \tag{19}$$

where N donates the number of input vectors and x_i and y_i are, respectively, the predicted vector and the truth vector; besides, $\sigma(x) = (1/(1 + e^{-x}))$.

IV. Experiment Analysis

The following section describes the experimental procedure and related details.

A. Experimental Data Description

1) Dataset:

a) DOTA-SHIP: The DOTA [40] dataset consists of 2806 aerial images acquired from different sensors and platforms with 15 common target classes. Image sizes in DOTA span from 800 × 800 to 4000 × 4000 pixels. We selected 420 images with 36 258 ship objects to compose the new ship dataset DOTA-SHIP. We randomly assigned the images in the dataset and obtained 315 and 105 images for the training set and the test set. Because the original images in DOTA are too big, we crop them to 1024 × 1024 and preserve as many ship objects as we can.

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TABLE I
EFFECTS OF EACH MODULE ON THE DOTA-SHIP DATASET

Model	PrfoIoU	CSP-COT	LTSCODE	AP50(%)	AP (%)	Parameters(M)	FPS
R-YOLOV5m				94.4	63.9	21.58	96.1
R-YOLOV5s				93.3	60.5	7.50	149.2
		√		94.7	61.2	7.48	126.6
			√	94.8	63.2	25.85	92.6
	√			94.5	61.1	7.40	156.1
	√	√	√	96.7	66.4	25.51	100.0

TABLE II
EFFECTS OF EACH MODULE ON THE UCAS-AOD DATASET

Model	PrfoIoU	CSP-COT	LTSCODE	AP50(%)	AP (%)	Parameters(M)	FPS
R-YOLOV5m				90.2	58.5	21.58	78.7
R-YOLOV5s				87.1	54.8	7.50	149.2
		√		89.3	55.6	7.48	147.1
			√	90.9	58.3	25.85	84.1
	√			88.9	55.4	7.40	156.1
	√	√	√	92.9	59.1	25.51	84.7

b) HRSC2016: The HRSC2016 [41] dataset consists of 1061 remote sensing images from six different ports. Ships at anchor and ships at sea are two examples of detecting objects in this dataset. The images can be divided into 436 training sets (a total of 1207 labeled examples), 444 test sets (a total of 1228 labeled examples), and 181 validation sets (a total of 541 labeled examples). The size of the image ranges from 300 × 300 to 1500 × 900 pixels.

c) UCAS-AOD: UCAS-AOD [42] is an aerial vehicle and automobile detection dataset containing 1510 images. The dataset is based on Google Earth, so the images all contain Google Earth watermarks. It consists of 1000 plane images and 510 car images annotated by directional bounding boxes. We randomly divided the training set and the verification set into 3:1. We use this dataset to verify the generalization of the model.

2) Experimental Details: The experimental environment was conducted with a deep learning framework based on Python's PyTorch 1.8.0, the test and validation environments are Windows 10 systems, and the evaluation of the model performance was performed on an 8-GB NVIDIA GeForce RTX 3070 GPU.

For the DOTA-SHIP dataset and the UCAS-AOD dataset, training was performed using the adaptive moment estimation (ADAM) optimizer [43] with the learning rate set to 0.001. The epoch parameter of the dataset is set to 300, and the batch size is set to 8. The stochastic gradient descent (SGD) algorithm [44] was used to train the HRSC2016 dataset, with a learning rate of 0.01. The decay rate of the learning rate is set to 0.2. The epoch parameter of the dataset is set to 500, and the batch size is set to 8. The momentum is set to 0.937, the weight_decay is set to 0.0005, and the warmup_epochs is set to 3. The test IOU threshold for all datasets is set to 0.4. To ensure consistency in training, other parameters are used by default. Feeding these high-resolution images directly into the network increases the computational burden and greatly increases the time loss. In order to facilitate comparison and reduce the calculation burden, we standardize the above datasets with a uniform input size of 800 × 800 pixels.

B. Ablation Experiment

1) Model Evaluation: To prove the superiority of the additional modules, we ran a number of experiments on the DOTA-SHIP dataset, the HRSC2016 dataset, and the UCAS-AOD dataset. Considering the parameter issues, we added the YOLOV5m model as the comparison model. Tables I–III show the experimental results.

Utilizing the experimental results obtained from the DOTA-SHIP dataset, we delve into a comprehensive analysis of the model's performance. Initially, the baseline YOLOv5 model exhibits a limited AP50 [5] of 34.4%, attributed to its inability to detect rotating targets. This underscores the challenges faced by the model in the rotating target detection task. We introduce an angle regression to overcome this limitation, used for angular prediction. After that, we observe a remarkable enhancement in performance. Specifically, the AP50 improves by a substantial 58.9%, while the overall AP also increases by 46.8%. Furthermore, we introduce the CSP-COT module to extract global spatial and contextual information from deep feature maps. This module not only effectively highlights crucial information but also reduces computational complexity and model parameters by replacing

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TABLE III
EFFECTS OF EACH MODULE ON THE HRSC2016 DATASET

Model	PrfoIoU	CSP-COT	LTSCODE	AP50(%)	Parameters(M)	FPS
R-YOLOV5m				87.6	21.58	71.3
R-YOLOV5s				86.4	7.50	121.9
		√		87.9	7.48	135.1
			√	88.4	25.85	75.7
	√			87.5	7.40	156.1
	√	√	√	91.0	25.51	75.8

TABLE IV
EFFECTS OF CSP BOTTLENECK AND CSP-COT ON THE DOTA-SHIP DATASET

Method	CSP Bottleneck	Vision Transformer	Bottleneck Transformers	CSP-COT	AP50(%)	AP (%)	Parameters(M)	FPS
R-YOLOV5m	√				94.4	63.9	21.58	96.1
R-YOLOV5S	√				93.3	60.5	7.50	149.2
		√			93.5	57.4	7.51	97.1
			√		94.0	61.1	7.10	95.2
				√	94.7	61.2	7.48	126.6

traditional 3 × 3 convolutions with MHSA. Experimental results reveal that on the DOTA-SHIP dataset, the inclusion of the CSP-COT module leads to a 1.4% increase in AP50 and a 0.7% boost in AP. In addition, on the UCAS-AOD and HRSC2016 datasets, the module significantly improves AP50 and AP while reducing model parameters by 0.2M, thus validating its effectiveness in optimizing model performance.

Moreover, the LTSCODE module enhances information fusion in feature maps by fusing different levels of feature maps according to the tasks of various feature layers. This approach achieves multifeature information interaction and decouples the tasks of angle prediction, localization, and categorization. Consequently, the LTSCODE module improves AP50 by 1.5% and AP by 2.7% on the DOTA-SHIP dataset, 3.8% and 3.5% on the UCAS-AOD dataset, and the AP by 2.0% on the HRSC2016 dataset, demonstrating its significance in improving model performance. However, it is worth noting that the design concept of LSTCODE is to increase the input of feature information to enhance the stable detection of global information by the detection head. Therefore, LSTCODE has a relatively large number of parameters, but we believe that this phenomenon is reasonable. Through Table I, we can find that there is a significant improvement in the performance of LSTCODE under both AP50 and AP evaluation metrics. This indicates that LSTCODE has stronger generalization ability, especially when dealing with complex scenes.

Finally, we also designed the PrfoIoU module to enhance the rotational accuracy of the model. The experimental results show that on the DOTA-SHIP dataset, the PrfoIoU module increases AP50 by 1.2% and AP by 0.6%. On the UCAS-AOD and HRSC2016 datasets, the PrfoIoU module significantly improves AP50 and AP while reducing model parameters by 0.4M, demonstrating its accuracy in angle regression.

By integrating the PrfoIoU, CSP-COT, and LTSCODE modules, we construct the CGTC-RYOLO model for rotating target detection. This model exhibits excellent performance on the DOTA-SHIP, HRSC2016, and UCAS-AOD datasets. Compared to the baseline rotational detection model, CGTC-RYOLO enhances AP50 by 3.4% on the DOTA-SHIP dataset, 4.8% on the HRSC2016 dataset, and 4.5% on the UCAS-AOD dataset. These results unequivocally demonstrate the accuracy and effectiveness of the CGTC-RYOLO model in rotating target detection tasks.

2) Evaluation of CSP-COT: In this section, we focus on comparing the effects of different CSP and MHSA modules on the ship detection performance, mainly including CSP bottleneck, vision transformer, bottleneck transformers [32], and CSP-COT. The original YOLOV5 structure uses the CSP bottleneck structure, and we replaced it with CSP-COT. The comparison effect of each model in different datasets is shown in Tables IV and V.

Comparison tests on the DOTA-SHIP dataset can show that compared to the traditional CSP bottleneck module, the CSP module that introduces the MHSA with convolutional mixing not only reduces the model size and parameters to a certain extent but also improves the AP value, which can prove that the MHSA is helpful for the detection of the ships in the remote sensing imagery. Specifically, the AP50 value of the model on the DOTA-SHIP dataset increased from 93.3% to 94.7%, the AP value increased from 60.5% to 61.2%, and the model parameters remained unchanged. The AP50 value of the model on the HRSC2016 datasets increased from 86.4% to 87.9%, the AP value increased from 57.8% to 63.0%, and the model parameters also remained unchanged.

However, from Tables IV and V, we can also see that the floating-point computational cost of CSP-COT is higher than

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TABLE V
EFFECTS OF CSP BOTTLENECK AND CSP-COT ON THE HRSC2016 DATASET

Method	CSP Bottleneck	Vision Transformer	Bottleneck Transformers	CSP-COT	AP50(%)	AP (%)	Parameters(M)	FPS
R-YOLOV5m	√				87.6	74.6	21.58	71.3
	√				86.4	57.8	7.50	156.2
R-YOLOV5S		√			86.8	59.7	7.51	140.8
			√		87.7	62.8	7.10	133.3
				√	87.9	63.0	7.48	135.1

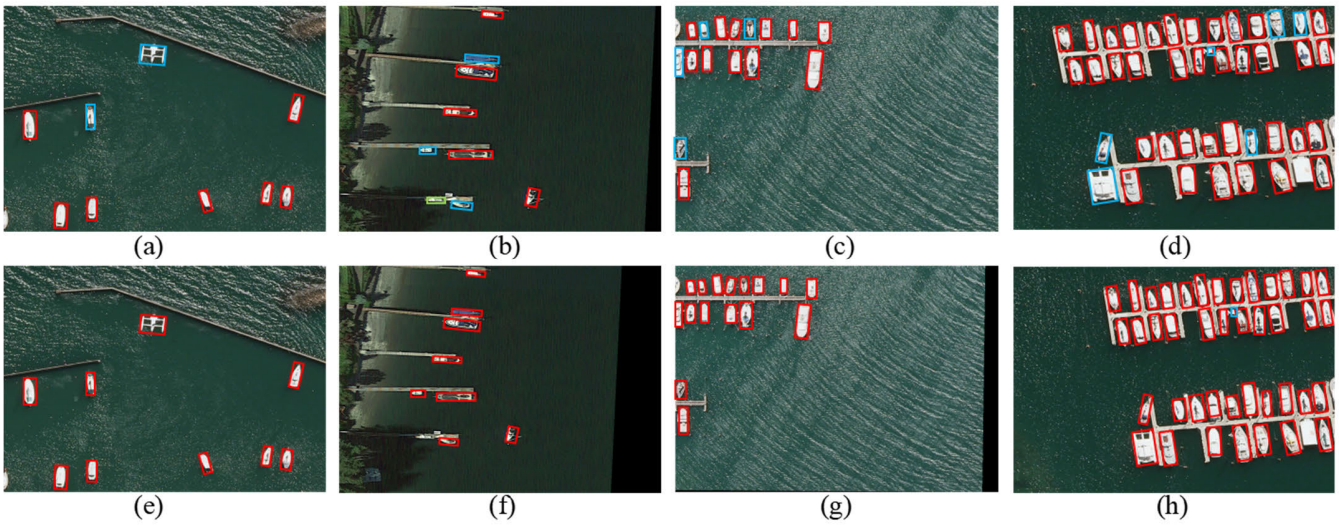


Fig. 4. Predicted OBBs' comparisons with/without COT. (a)–(d) Baseline. (e)–(h) COT. The red rectangle is the detection result, and the blue rectangle and green oval represent the OBB of missed and false positives, respectively.

that of CSP. This is because the CSP-COT module introduces more feature fusion and information integration mechanisms in its design to improve detection accuracy. This complexity naturally leads to an increase in computational costs. With the improvement of model capabilities, the increase in computational costs is a reasonable result. Although CSP-COT has increased in computational cost, it significantly improves the accuracy and reliability of rotation detection through more effective information integration and feature extraction. This performance improvement is crucial in many application scenarios, especially for object detection tasks in complex environments. As shown in Fig. 4, the detection performance of CSP-COT is significantly better in complex scenes.

Overall, the CSP-COT module is more suitable for ship inspection than the CSP bottleneck module. Convolutional and MHSA mechanisms are used in the hybrid model CSP-COT, which uses contextual information between input keys to guide self-attention learning. The COT module first captures static contextual knowledge between neighboring keys and then uses this information to further mine and enhance dynamic context self-attention. This method integrates self-attention learning and context mining into a single architecture, improving the visual representation. The reason is that using the MHSA module instead of the 3×3 convolution in the traditional CSP bottleneck module has the following advantages. First, the MHSA can consider global and valid information to encode contextual cues through a flexible and dynamic receptive field, thus refining the drawbacks of the original use of a fixed 3×3 convolution that focuses on only a small and fixed receptive field. Second, in comparison to the original CNN model in this task, the model that includes the MHSA can significantly improve the detection accuracy. Finally, MHSA has fewer parameters and additional floating-point operations compared to convolution. The visualization results are shown in Fig. 4. CSP-COT can work in the case of a dense arrangement of targets with only one missed target detection. Compared to the baseline, it can also utilize the acquired global context information to achieve effective identification of global targets without misdetected targets.

From Fig. 5, it can be observed that the CSP-COT module performs well in integrating global information. Specifically, the CSP-COT module optimizes the network's utilization of global contextual information, effectively combining shallow positional information with deep semantic information. This fusion method enables the model to not only obtain the precise location of the target but also understand the semantic features of the target, thus more comprehensively capturing the shape and characteristics of the ship. By combining deep and shallow information, the CSP-COT module enhances the network's understanding of the surrounding environment, enhances its ability to express target features, and makes the generation of detection boxes more in line with actual targets. This accurate framework enables the model to maintain efficient detection performance in complex scenes, which is

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Fig. 5. Predicted OBBs' comparisons by applying the COT and TSCODE + COT. (a) Baseline. (b) COT. (c) TSCODE + COT.

TABLE VI
DECOUPLED HEADERS ON DOTA-SHIP

Detect Head	AP50(%)	AP (%)	Parameters(M)
R-YOLOV5s	93.3	60.5	7.50
R-YOLOV5m	94.4	63.9	21.50
R-YOLOVX	93.8	62.9	20.24
R-YOLOV6	93.7	62.1	14.04
R-YOLOV8	92.6	60.0	37.37
R-LTSCODE	94.8	63.2	25.85
R-BOT-LTSCODE	95.4	63.6	25.12
R-COT-LTSCODE	95.4	64.6	25.51

particularly important for processing remote sensing images with complex backgrounds and diverse targets, helping to reduce false positives and false negatives. Overall, the CSP-COT module not only improves the accuracy of rotation detection but also enhances the overall detection efficiency, providing more reliable technical support for practical applications.

We also conducted a scatter plot comparison of the superiority of CSP-COT on the DOTA-SHIP and HRSC2016 datasets, as shown in Fig. 6. It can be clearly seen from the graph that the scatter plot of CSP-COT detection results shows higher density in both datasets, and its performance is superior compared to the CSP bottleneck. This result not only highlights the effectiveness of CSP-COT in object detection tasks but also demonstrates its reliability and accuracy in handling complex scenes, further demonstrating the significant advantages of CSP-COT in improving detection performance.

3) Evaluation of LTSCODE: LTSCODE decouples the feature encoding of the two tasks in the source code and utilizes feature maps with different semantic contexts in the two branches. The LTSCODE ablation comparison experiments are shown in Table VI. We grafted several classical decoupled head structures onto YOLOV5 and added the angle regression, resulting in the decoupled headers of YOLOV5m, YOLOVX, and YOLOV6 with rotation detection effect as our comparison model. The most primitive YOLOV5 series uses a 1×1 convolution for classification and localization regression in order to minimize performance loss and reduce the number of parameters. YOLOVX and YOLOV6 have a similar structure of the decoupling head, except that they reduce the number of convolutions by a 3×3 convolution on the classification and localization branches. Comparison experiments on the DOTA-SHIP dataset show that YOLOV6 increases the AP50 from 93.3% to 93.7% by adding the decoupling head that separates the

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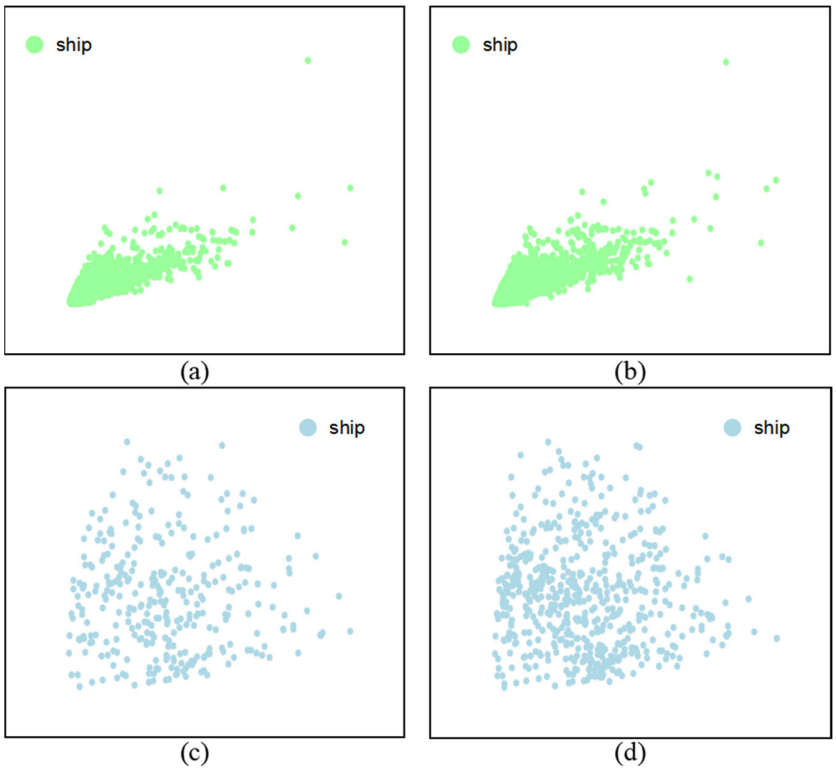


Fig. 6. Scatter plot comparisons with/without COT. (a) Scatter plot of CSP bottleneck in the DOTA-SHIP dataset. (b) Scatter plot of CSP-COT in the DOTA-SHIP dataset. (c) Scatter plot of CSP bottleneck in the HRSC2016 dataset. (d) Scatter plot of CSP-COT in the HRSC2016 dataset.

TABLE VII
LOSS FUNCTIONS ON DOTA-SHIP

LOSS	AP50(%)	AP (%)
Smooth-L1	82.2	43.7
CSL	84.3	48.1
KLD	88.3	57.6
GWD	85.7	52.9
PrfoIoU	94.5	61.1

classification and localization tasks. YOLOVX improves AP50 from 93.3% to 93.8% by increasing the number of convolutional layers, and finally, our LTSCODE achieves a significant improvement of 1.5% in AP50 and 2.7% in AP, which shows that LTSCODE can further fuse the multilevel feature maps to effectively resolve the contradictory conflicts between localization and classification and help the detector accurately to localize rotating ships with large size variations.

Then, we introduce the current popular CSP bottleneck transformer (CSP-BOT) [32] as a comparison. CSP-BOT is also a hybrid model using convolution and MHSA, which can change the weight of each pixel in the feature map based on the global attention in several subspaces and focus on the target and the important regions near the target. Compared with CSP-BOT, CSP-COT pays more attention to static and dynamic context information. The BOT-LTSCODE, which incorporates the attention mechanism, achieves a 2.1% improvement on AP50 and 3.1% on AP relative to the baseline model. Finally, our COT-LTSCODE improved by 2.1% on AP50 and 4.1% on AP. As show in Predicted results at this time, the COT-LTSCODE is more accurate than baseline and BOT.

4) Evaluation of PrfoIoU: According to Table VII, we also tested the superiority of the PrfoIoU module. We have selected four currently advanced rotational losses whose backbones are all YOLOV5S for comparison, respectively: Smooth-L1 [59], CSL, KLD, and GWD. The experimental results show that the performance of ProIoU loss proposed in this article is superior to other methods, with improved AP50 and AP values.

C. Comparisons With State-of-the-Art Models

In this section, we chose some mainstream one- and two-stage detection methods for comparison, such as R2CNN [46], SLA [22], CFCNet [2], and RoI transformer [24]. These methods have strong reference value until now.

1) DOTA-SHIP: In order to visualize the test results for comparison while saving space, we only present the test results for the four methods, as shown in Fig. 7. Detailed results for other models are shown in Table VIII.

Table VIII shows the comparison of the different methods on the DOTA-SHIP dataset in terms of AP50, parameters, and model sizes. Based on the MS COCO evaluation metrics [41], we can obtain relevant results. It can be seen that the traditional

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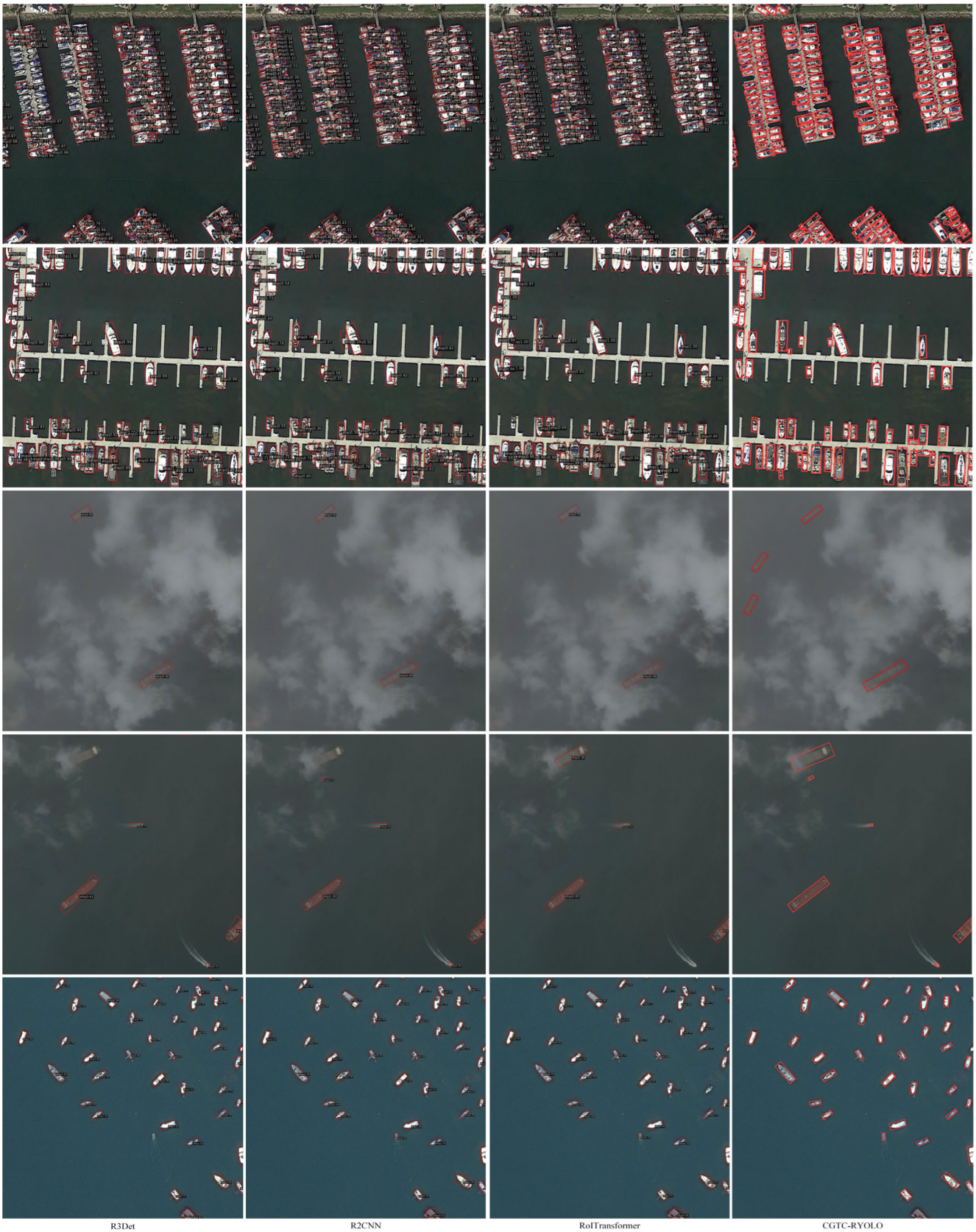


Fig. 7. Detection results on the DOTA-SHIP dataset. The visualization results of four representative methods are obtained in typical and complex scenes, such as dense arrangement, cloud occlusion, and sparse arrangement, respectively.

two-stage model has a large number of parameters, but the detection accuracy is lower than the one-stage model, which is not suitable for oriented ship detection under dense arrangement. Among all the compared methods, our method

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TABLE VIII
COMPARISON OF OTHER METHODS AND OUR METHODS ON THE DOTA-SHIP DATASET

Algorithms	Method	Back bone	Input size	AP50(%)	Parameters(M)	Model Size(M)
Two-stage	R2CNN [46]	ResNet50	800×800	56.3	41.1	338.3
	RRPN [21]	ResNet50	800×800	59.1	36.6	213.2
	CSL [23]	ResNet-50	800×800	71.9	41.1	314.0
	RoI Transformer [24]	ResNet101	800×800	80.4	55.1	421.1
One-stage	R3Det [47]	ResNet101	800×800	60.1	41.5	318.6
	S2ANet [48]	ResNet50	800×800	71.2	38.5	295.3
	SLA [22]	ResNet-50	800×800	88.4	40.9	293.1
	SD-Net [49]	ResNet-101	800×800	91.4	42.0	-
	YOLOV7 [50]	ELAN-Net	800×800	93.8	34.3	71.8
	YOLOV8 [51]	CSP-DarkNet	800×800	92.6	37.2	25.5
	CGTC-RYOLO	YOLOv5s	800×800	96.7	25.5	50.1

TABLE IX
COMPARISON OF OTHER METHODS AND OUR METHODS ON THE HRSC2016 DATASET

Algorithms	Method	Back bone	Input size	AP50(VOC2007)	Parameters(M)	Model Size(M)
Two-stage	R2CNN [46]	ResNet-50	800×800	73.1	41.1	314.0
	RRPN [21]	ResNet-50	800×800	79.6	36.6	213.2
	CSL [23]	ResNet-50	800×800	89.6	37.7	286.3
	R3Det [47]	ResNet-101	800×800	89.3	41.5	318.6
	RoI Transformer [24]	ResNet-101	512×800	86.2	55.1	421.1
	OrientedRCNN [51]	ResNet-101	1333×800	90.5	41.1	158.1
	Gliding Vertex [52]	ResNet101	512×800	88.2	-	-
One-stage	S2ANet [48]	ResNet-50	800×800	90.2	38.5	295.3
	SLA [22]	ResNet-50	800×800	87.1	40.8	293.1
	CFCNet [2]	ResNet-50	800×800	89.7	37.5	450.9
	SD-Net [57]	ResNet-101	800×800	90.3	42.0	-
	RIDet-Q [53]	ResNet-50	800×800	89.6	37.7	431.0
	RIDet-O [53]	ResNet-50	800×800	89.1	37.7	431.0
	RSDet++ [54]	ResNet-152	800×800	86.5	-	-
	OPLD [55]	ResNet-50	1024×1333	88.4	-	-
	OSSDet [56]	ResNet-50	800×800	89.8	-	-
	YOLOV6 [11]	EfficientRep	1024×1024	86.1	34.9	72.6
	YOLOV8	CSP-DarkNet	1024×1024	85.7	37.2	25.5
	CGTC-RYOLO	YOLOv5s	800×800	91.0	25.5	50.1

has the highest detection accuracy of 96.7% for AP50. Compared to YOLOV7 and YOLOV8, which are similar research methods, the detection accuracy increased by 2.9% and 4.1%, respectively.

Visualizations of some of these detections are shown in Fig. 7. We can see that in the case of a dense array of targets, the traditional method will almost produce a certain amount of missing cases. This is because the global relationship information extracted by the network is not sufficient, resulting in that the dense array of targets cannot be accurately detected. In a complex scene with cloud occlusion, the traditional method cannot obtain the complete semantic information of the occluded object, so there is a case of missing detection. However, in the scenario of sparse targets, only one or two targets are missed by the traditional method, which achieves good detection performance. CGTC-RYOLO can better identify each ship in complex scenarios in these complex situations and obtain high-quality rotation detection results to achieve a high detection rate.

2) HRSC2016: Table IX shows the comparison of the different methods on the HRSC2016 dataset in terms of AP50 and model size. Based on the MS COCO assessment metrics, we can obtain relevant results. OrientedRCNN has the highest detection accuracy in the two-stage detection algorithm with 90.5%, which is only 0.5% lower than our algorithm. Our CGTC-RYOLO achieved the highest performance in the one-stage algorithm with an accuracy of 91.0% for AP50.

Similarly, we have chosen representative single-stage object detection algorithms such as YOLOv7 and YOLOv8 for comparison, and we can see that our method is superior to them. This shows that traditional HBB algorithms such as the YOLO series are not applicable to the HRSC2016 dataset. We visualize some of the detection results, as shown in Fig. 8, where our model can accurately label the ships.

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TABLE X
COMPARISON OF OTHER METHODS AND OUR METHODS ON THE UCAS-AOD DATASET

Method	Back bone	Car	Plane	AP50(%)	Parameters(M)	Model Size(M)
R2CNN [46]	ResNet-50	86.8	89.8	88.3	41.4	314.0
RoI Transformer [24]	ResNet-50	88.0	90.0	89.0	55.3	421.1
R-YOLOV3 [8]	DarkNet53	74.6	89.5	82.0	61.6	704.7
R-RetinaNet [37]	ResNet-50	84.6	90.5	87.5	36.3	139.1
SLA [22]	ResNet-50	88.5	90.3	89.4	40.9	293.1
CFCNet [2]	ResNet-50	89.2	88.6	89.4	37.5	450.9
SD-Net [49]	ResNet-101	-	-	90.2	42.0	-
RIDet-Q [53]	ResNet-50	88.0	89.9	89.2	37.7	431.0
RIDet-O [53]	ResNet-50	88.8	90.3	89.6	37.7	431.0
Yu et al. [57]	ResNet-50	88.9	90.4	89.7	-	-
Yu et al. [58]	ResNet-101	86.6	94.2	90.5	-	-
CGTC-RYOLO	YOLOV5s	89.5	96.1	92.9	25.5	50.1



Fig. 8. Detection results on the HRSC2016 dataset.

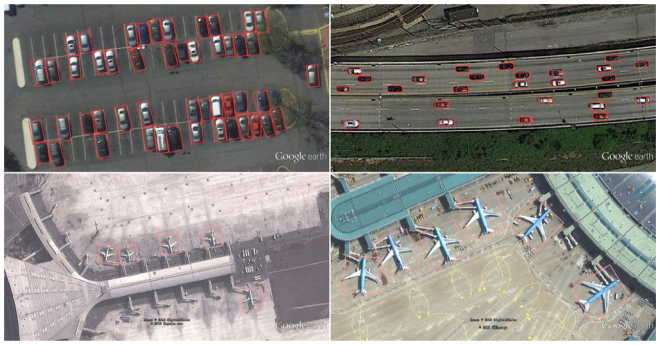


Fig. 9. Detection results on the UCAS-AOD dataset.

3) UCAS-AOD: Some vehicles and aircraft in the UCAS-AOD dataset have different orientations, significant size differences, and dense distributions, which are difficult to detect, so this dataset is taken in this experiment to verify the generalization of the model. Table X shows the comparison of model parameters, model sizes, and AP50 for aircraft and automobiles on UCAS-AOD datasets by different methods. Among the compared methods, our method has the highest detection accuracy of 92.9% for AP50. In addition, the model only needs 50.1 MB of storage space, which is several times smaller than the other variants [48], [60], [61], [62], [63], [64], [65]. Fig. 9 shows some of the visualization results from the UCAS-AOD dataset. Our new framework can accurately identify each vehicle and aircraft in complex background conditions, such as city roads and airports, and output high-quality rotating bounding boxes. This also confirms the effectiveness of our PrfoIoU and CSP-COT modules in angle processing in complex scenes. A series of experimental results on the UCAS-AOD dataset also demonstrate the generalizability of our model.

V. CONCLUSION

In this article, a new network CGTC-RYOLO for rotating frame detection based on YOLOV5 is proposed. The key improvements are the cross-stage local information fusion module and the construction of a new decoupling head framework. The CSP module is replaced by the CSP-COT module using a mixture of convolutional and MHSA mechanisms, thereby capturing global information about the target distribution and enhancing critical information. The use of LTSCODE, which adaptively aggregates features of different resolutions through weighted cross-scale connections to accommodate spatial variations in prediction uncertainty, is used to resolve the contradiction between YOLOV5 decoupled head classification and localization.

In addition, the importance and effectiveness of each module are demonstrated by ablation experiments. After that, we compare the performance of the CGTC-RYOLO network with other popular detection networks, and extensive experiments on the DOTA-SHIP dataset, the HRSC2016 dataset, and the UCAS-AOD dataset show that the proposed method can all achieve better accuracy. Overall, the method achieves excellent ship detection performance among the mainstream detection methods for rotating target detection in remote sensing images.

The method proposed in this article is mainly aimed at ship targets. Although other types of targets were also involved in the experiment, due to the limitation of the research scope, we only conducted preliminary discussions. We hope that this can provide a starting point for interested researchers and encourage them to conduct more in-depth exploration and

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research in related fields. In addition, in order to improve the detection performance of the model, the design of LSTCODE includes many parameters. Although this may increase the complexity of the model, it also lays an important foundation for performance improvement. We encourage readers to draw inspiration from it and actively explore new ideas that can effectively reduce the number of parameters while improving accuracy in order to achieve better model efficiency. We hope that our research can inspire readers to design more efficient and universal network architectures, thereby promoting further development and progress in the field of object detection. Meanwhile, we look forward to future research based on our work, exploring more practical applications to meet the growing technological demands.

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